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A HYBRID APPROACH TO PSYCHIATRIC SCREENING CLASSIFICATION BASED ON EEG SPECTRAL FEATURES

Abstract*:* this study continues an exploratory analysis of EEG spectral features by comparing supervised classifiers and two combinations of methods. L1-based feature selection and autoencoder representations were paired with Random Forest. Evaluation used separate fitting, calibration and test partitions, with thresholds targeting sensitivity of at least 0.90 on calibration data. The combinations did not establish a consistent advantage over individual classifiers. Most healthy controls still received positive screening predictions, defining a limitation for further development.

Keywords*:* electroencephalography, spectral features, L1-based feature selection, autoencoder, Random Forest, psychiatric screening.

Introduction

The previous study described spectral differences and examined whether deviations from a healthy reference could distinguish patients from healthy controls [1]. Both initial anomaly detectors flagged most healthy participants. This raised a further question: could supervised classification make better use of the same spectral features? The present work compares individual classifiers with two hybrid approaches that combine feature selection or learned representations with Random Forest. The aim is to determine whether these combinations improve discrimination between patients and healthy controls while prioritizing patient detection. The task

remains binary and does not involve identifying individual psychiatric disorders.

Dataset and hybrid approaches

The same open EEG Psychiatric Disorders Dataset was used [2]. Restricting the analysis to ages 18–30 retained 599 participants: 515 patients and 84 healthy controls. Inputs were 114 pre-calculated absolute-power features: six frequency bands at 19 electrodes. Diagnoses were combined into one patient class; individual disorders were not predicted. Coherence variables and demographic characteristics were excluded from model inputs.

L1 logistic regression was used both as a classifier and to rank original features by absolute coefficient magnitude. The L1 penalty can produce sparse coefficients [3]; the 20 highest-ranked features were passed to Random Forest. This tests whether reducing the original feature set helps an ensemble that can capture nonlinear relationships [4]. A forest using all 114 features provided the comparison. In four outer folds, fewer than 20 coefficients were nonzero, so zero-coefficient features completed the fixed-size set.

The second approach used an autoencoder to learn a compact representation through reconstruction [5]. Its layer sizes were 114–64–32–64–114, with ReLU in hidden layers and a linear output. Training used Adam, mean squared reconstruction error, 50 epochs and batches of 32. The 32 bottleneck values became inputs to Random Forest. Unlike the earlier anomaly detector, this autoencoder learned from both groups in the fitting partition without using their labels. The two feature-processing paths are summarized in Figure 1.

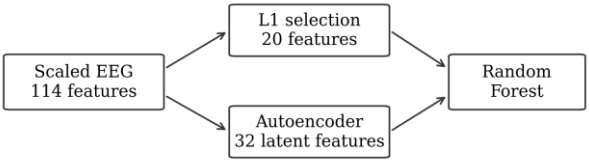


Figure 1 – Two feature-processing paths for Random Forest

A most-frequent-class Dummy classifier completed the comparison. It predicts every participant as a patient, providing a baseline for the strongly imbalanced sample.

Evaluation and screening thresholds

Five outer folds were evaluated in two repeats. Each outer training partition was further divided into fitting and calibration subsets, giving approximately 60%, 20% and 20% of the full sample for fitting, calibration and testing. All five methods used identical partitions. Records with identical spectral features were grouped during splitting to prevent data leakage.

Hyperparameters were selected by three-fold grouped cross-validation within the fitting subset, using balanced accuracy. Standardisation, L1 selection and autoencoder training were repeated inside each internal training fold. Logistic regression used C values of 0.1, 1 and 10. All forest variants used 200 trees, balanced class weights, maximum depth of 6 or unrestricted depth, and minimum leaf sizes of 1 or 3. The autoencoder architecture was fixed rather than tuned.

Each selected pipeline was trained on the fitting subset. A decision threshold was then selected using a separate calibration subset to correctly exclude as many healthy controls as possible while identifying at least 90% of patients:

$$\text{Sensitivity}_{\text{cal}}(\tau^*) \geq 0.90, \quad (1)$$

where

Sensitivity– the proportion of patients correctly identified in the calibration subset;

cal – the calibration subset.

τ^* – the selected decision threshold.

Scores at or above the threshold indicated the patient class. If several thresholds achieved the same highest specificity under condition (1), the highest threshold was selected. This target was experimental, not a clinical standard.

Sensitivity measures patient detection, while specificity measures correct recognition of healthy controls. Balanced accuracy averages these two rates. ROC AUC measures how well model scores distinguish the groups across thresholds. Results are reported as means and standard deviations across ten outer evaluations. Since these evaluations reuse participants, the standard deviations describe variation across splits and are not confidence intervals.

Results and interpretation

L1 selection with Random Forest had the highest mean specificity and balanced accuracy in the calibrated screening regime (Figure 2). Its sensitivity was 0.896 and specificity 0.208. Autoencoder features with Random Forest produced similar values, 0.892 and 0.201. The differences between the learned models were small relative to the variation across folds; these results do not establish statistical superiority.

Model	Sensitivity	Specificity	Balanced accuracy	ROC AUC
Dummy classifier	1.000 ± 0.000	0.000 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
L1 logistic regression	0.884 ± 0.043	0.203 ± 0.122	0.543 ± 0.068	0.658 ± 0.057
Random Forest	0.890 ± 0.050	0.191 ± 0.128	0.541 ± 0.066	0.632 ± 0.058
L1 selection + Random Forest	0.896 ± 0.049	0.208 ± 0.101	0.552 ± 0.030	0.637 ± 0.052
Autoencoder + Random Forest	0.892 ± 0.055	0.201 ± 0.157	0.547 ± 0.062	0.620 ± 0.043

Figure 2 – Test performance with calibration sensitivity target ≥ 0.90

Values are mean ± SD. L1+RF denotes selection of 20 original features; AE+RF denotes classification of 32 latent features. Individual L1 had the highest mean ROC AUC (0.658), whereas the hybrid approaches did not improve ranking.

Calibration changed the balance of errors. For L1-RF, standard prediction gave sensitivity 0.937 and specificity 0.119; the calibrated rule reduced sensitivity to 0.896 while increasing specificity to 0.208. For AE-RF, the corresponding changes were 0.952 to 0.892 and 0.089 to 0.201. Thus, threshold selection excluded more healthy participants from positive screening, at the cost of missing more patients. Dummy detected every patient but excluded no healthy controls.

Discussion and conclusion

Supervised classification provided some ability to distinguish patients from healthy controls, but neither hybrid approach demonstrated a clear advantage over individual classifiers. At the calibrated thresholds, the models correctly excluded approximately one fifth of healthy controls while detecting about nine tenths of patients. Compared with referring everyone, this reduced unnecessary referrals at the cost of missed patients. However, these results do not yet support practical screening use.

Interpretation is limited by the small control group, heterogeneous diagnoses and evaluation on previously explored data. Selected features and latent representations were not validated as biomarkers. Further research is warranted to determine whether additional EEG characteristics, including coherence, can improve specificity while maintaining high sensitivity. Any improvement would then require validation on independent data.

References

[1] Evseeva O., Yafremava A.Y. Exploratory analysis of EEG spectral features for psychiatric screening // The Priority Research Areas in the XXI Century: conference proceedings / ed. A.I. Vostretsov. – Neftekamsk: Mir nauki, 2026. – P. 14–19.

[2] Park S.M., Jeong B., Oh D.Y. et al. Identification of Major Psychiatric Disorders From Resting-State Electroencephalography Using a Machine Learning Approach // Frontiers in Psychiatry. – 2021. – Vol. 12. – Article 707581. – DOI: 10.3389/fpsyt.2021.707581.

[3] Tibshirani R. Regression Shrinkage and Selection Via the Lasso // Journal of the Royal Statistical Society: Series B (Methodological). – 1996. – Vol. 58, No. 1. – P. 267–288. – DOI: 10.1111/j.2517-6161.1996.tb02080.x.

[4] Breiman L. Random Forests // Machine Learning. – 2001. – Vol. 45. – P. 5–32. – DOI: 10.1023/A:1010933404324.

[5] Hinton G.E., Salakhutdinov R.R. Reducing the Dimensionality of Data with Neural Networks // Science. – 2006. – Vol. 313, No. 5786. – P. 504–507. – DOI: 10.1126/science.1127647.

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